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MID SEMESTER RE EXAMINATION 2025-26Name of the Course: B. Tech. Biotechnology

Semester: V

Name of the Paper: Fundamentals of AI and MLPaper Code: TBT509b

Time: 1:30 Hrs

Maximum Marks: 50

Note: -

- (i) All questions are compulsory.
- (ii) Do any one of the (a) or (b) in each question
- (iii) Each question carries 10 marks

Q1	(10marks)	CO1,CO2
(a)	Artificial Intelligence is often described as the simulation of human cognitive processes. Discuss the various components of AI that mimic human intelligence such as learning, reasoning, perception, and language understanding. How does each component contribute to the creation of intelligent systems like self-driving cars and virtual assistants?	
	OR	
(b)	Consider the historical evolution of AI from the 1950s to the present. How have shifts in AI paradigms (symbolic AI to machine learning to deep learning) influenced the type of problems AI can solve today? Provide an analysis of how these shifts have led to breakthroughs in fields like healthcare, finance, and robotics.	
Q2	(10marks)	CO1, CO2
(a)	Define and compare the different types of Machine Learning: Supervised Learning, Unsupervised Learning, and Reinforcement Learning. Provide practical examples for each type, and explain how they differ in terms of data processing, learning method, and application.	
	OR	
(b)	Supervised learning relies on labeled data, but how does the choice of features in the training data influence the accuracy and robustness of a machine learning model? Critically discuss feature selection techniques such as filtering, wrapper methods, and embedded methods, and their significance in improving model performance.	
Q3	(10marks)	CO1, CO3
(a)	(i) Elaborate on the distinction between Narrow AI and General AI. How close are we to achieving General AI, and what would be the potential impact on industries like healthcare, education, and finance? (ii) A dataset of house prices contains the following features:	

Size (sq. ft.)	Price (lakhs)
1000	50
1500	65
2000	80
2500	95
3000	110

Derive the regression equation. Predict the price of a house with Size = 2200 sq. ft.

OR

- (b) What are the steps involved in the Data Mining process? Discuss each step (data collection, data cleaning, integration, transformation, reduction, mining, evaluation, and presentation) in detail with examples. How do these steps ensure the successful extraction of meaningful patterns?

Q4 (10 marks)

- (a) Explain the role of data mining techniques such as classification, clustering, regression, and association in extracting insights from data. Provide specific examples of how these techniques are applied in business scenarios like customer segmentation, product recommendations, or fraud detection.

**CO1, CO2,
CO5**

OR

- (b) Differentiate between databases and data warehouses. Discuss their roles in data management, their architecture, and how they are used in day-to-day operations versus long-term business decision-making. Include examples such as banking and e-commerce systems.

Q5 (10 marks)

- (a) Describe the ETL process (Extract, Transform, Load) in detail. How does this process facilitate the movement of data from operational databases to a data warehouse? Provide an example, such as Amazon's use of the ETL process in its sales data analysis.

**CO1, CO2,
CO5**

OR

(b)	What is information interpretation in data mining? Discuss how raw patterns discovered in data mining are evaluated, validated, and interpreted to make actionable business decisions. Use examples such as retail sales optimization or customer behavior analysis.
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